PROPER HYPERCOVERINGS: A^{-1} ON ETALE SHEAVES IS FULLY FAITHFUL WITH ESSENTIAL IMAGE THE CARTESIAN SHEAVES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-spaces-simplicial/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the fully-faithful half is obtained by what the author calls 'a formal argument (chasing around the diagram)' (source line 8065) through the comparison diagram of lemma-compare-simplicial-objects-ph-etale - all inputs (full faithfulness of h^{-1} and of a_{ph}^{-1} with quasi-inverses, the commutativities established in the cited lemmas) are in place before the sentence, and the step is glue between them; the essential-image half is carried by the complete six-equality computation, each equality justified by a named lemma in-span.

Editorial note. The label lemma-descent-sheaves-for-ph-hypercovering names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Comparison of the small etale and big ph simplicial sites. Same chapter, \label{lemma-compare-simplicial-objects-ph-etale}. Lines 8046-8048: the diagram h with h_{-1} = a_X and components h_n; the fully-faithfulness transfer chases this diagram.
- The ph-etale comparison lemma. More on Cohomology of Spaces chapter, \label{spaces-more-cohomology-lemma-comparison-ph-etale}. Lines 8049-8052: (h_{-1})^{-1} is fully faithful with quasi-inverse h_{-1,*}; the same holds for the components h_n.
- Description of the simplicial-site functor pair h^{-1}, h_*. Same chapter, \label{lemma-morphism-simplicial-sites}. Lines 8053-8054: componentwise full faithfulness lifts to h^{-1} fully faithful with quasi-inverse h_*.
- Simple sheaf descent for hypercoverings on the big ph site. Same chapter, \label{lemma-hypercovering-X-simple-descent-sheaves}. Lines 8061-8064: a_{ph}^{-1} fully faithful with quasi-inverse a_{ph,*} and essential image the cartesian sheaves on the big total site.

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- Proper pushforward and pullback along the ph-augmentation. More on Cohomology of Spaces chapter, \label{spaces-more-cohomology-lemma-proper-push-pull-ph-etale}. Lines 8097-8102: the third equality of the equalizer computation, using that a_0 and a_1 are proper.
- The augmentation site computes a_* as an equalizer. Same chapter, \label{lemma-augmentation-site}. Lines 8085-8096: the first and fifth equalities of the computation.
- Surjective proper morphisms are ph coverings. Topologies on Spaces chapter, \label{spaces-topologies-lemma-surjective-proper-ph}. Lines 8058-8061: U is a hypercovering of X in (Spaces/S)_ph.

Source reference. Original reference 0D70: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 1141–1141. Prerequisite pointer invoked at proof line(s) 8117; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0DHJ: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 7973–7973. Prerequisite pointer invoked at proof line(s) 8046; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0DA8: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 5065–5065. Prerequisite pointer invoked at proof line(s) 8063; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0D96: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 1273–1273. Prerequisite pointer invoked at proof line(s) 8057; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 09A0: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 5012–5012. Prerequisite pointer invoked at proof line(s) 8060; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 0DE7: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 6985–6985. Prerequisite pointer invoked at proof line(s) 8039; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0DGT: sources/stacks-chow/tags.txt, source lines 16883–16883. Prerequisite pointer invoked at proof line(s) 8053; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0DGX: sources/stacks-chow/tags.txt, source lines 16887–16887. Prerequisite pointer invoked at proof line(s) 8121; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 0DGS: sources/stacks-chow/tags.txt, source lines 16882–16882. Prerequisite pointer invoked at proof line(s) 8050; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0DFI: sources/stacks-chow/tags.txt, source lines 16838–16838. Prerequisite pointer invoked at proof line(s) 8062; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let S be a scheme. Let X be an algebraic space over S. Let U be a simplicial algebraic space over S. Let $a : U \to X$ be an augmentation. If $a : U \to X$ is a proper hypercovering of X, then*

$$a^{-1} : Sh(X_{étale}) \to Sh(U_{étale}) \quad \text{and} \quad a^{-1} : Ab(X_{étale}) \to Ab(U_{étale})$$

are fully faithful with essential image the cartesian sheaves and quasi-inverse given by a_*. Here $a : Sh(U_{étale}) \to Sh(X_{étale})$ is as in Section 0DE7.

Proof. We will prove the statement for sheaves of sets. It will be an almost formal consequence of results already established. Consider the diagram of Lemma 0DHJ. In the proof of this lemma we have seen that h_{-1} is the morphism a_X of More on Cohomology of Spaces, Section 0DGS. Thus it follows from More on Cohomology of Spaces, Lemma 0DGT that $(h_{-1})^{-1}$ is fully faithful with quasi-inverse $h_{-1,*}$. The same holds true for the components h_n of h. By the description of the functors h^{-1} and h_* of Lemma 0D96 we conclude that h^{-1} is fully faithful with quasi-inverse h_*. Observe that U is a hypercovering of X in $(Spaces/S)_{ph}$ as defined in Section 09A0 since a surjective proper morphism gives a ph covering by Topologies on Spaces, Lemma 0DFI. By Lemma 0DA8 we see that a_{ph}^{-1} is fully faithful with quasi-inverse $a_{ph,*}$ and with essential image the cartesian sheaves on $(Spaces/U)_{ph,total}$. A formal argument (chasing around the diagram) now shows that a^{-1} is fully faithful.

Finally, suppose that $\mathcal{G}$ is a cartesian sheaf on $U_{étale}$. Then $h^{-1}\mathcal{G}$ is a cartesian sheaf on $(Spaces/U)_{ph,total}$. Hence $h^{-1}\mathcal{G} = a_{ph}^{-1}\mathcal{H}$ for some sheaf $\mathcal{H}$ on $(Spaces/X)_{ph}$. We compute using somewhat pedantic notation

$$\begin{aligned}
(h_{-1})^{-1}(a_*\mathcal{G}) &= (h_{-1})^{-1}\mathrm{Eq}(\ a_{0,small,*}\mathcal{G}_0 \tightarrow a_{1,small,*}\mathcal{G}_1\ ) \\
&= \mathrm{Eq}(\ (h_{-1})^{-1}a_{0,small,*}\mathcal{G}_0 \tightarrow (h_{-1})^{-1}a_{1,small,*}\mathcal{G}_1\ ) \\
&= \mathrm{Eq}(\ a_{0,big,ph,*}h_0^{-1}\mathcal{G}_0 \tightarrow a_{1,big,ph,*}h_1^{-1}\mathcal{G}_1\ ) \\
&= \mathrm{Eq}(\ a_{0,big,ph,*}(a_{0,big,ph})^{-1}\mathcal{H} \tightarrow a_{1,big,ph,*}(a_{1,big,ph})^{-1}\mathcal{H}\ ) \\
&= a_{ph,*}a_{ph}^{-1}\mathcal{H} \\
&= \mathcal{H}
\end{aligned}$$

Here the first equality follows from Lemma 0D70, the second equality follows as $(h_{-1})^{-1}$ is an exact functor, the third equality follows from More on Cohomology of

Spaces, Lemma 0DGX (here we use that $a_0 : U_0 \to X$ and $a_1 : U_1 \to X$ are proper), the fourth follows from $a_{ph}^{-1}\mathcal{H} = h^{-1}\mathcal{G}$, the fifth from Lemma 0D70, and the sixth we've seen above. Since $a_{ph}^{-1}\mathcal{H} = h^{-1}\mathcal{G}$ we deduce that $h^{-1}\mathcal{G} \cong h^{-1}a^{-1}a_*\mathcal{G}$ which ends the proof by fully faithfulness of h^{-1}. $\square$